Multilayer Perceptron's in Practice: Investigating the Impact of Hidden Layer Depth on Fashion-MNIST Classification

GitHub: https://github.com/Deekshitha-1010/Machine-Learning/tree/main

1. Introduction

Artificial neural networks (ANN) is one of the most used machine learning techniques which helps in the solving of the various complex pattern recognition issues. One of the earliest and most basic neural networks used for supervised learning, such as classification or regression, is the Multilayer Perceptron (MLP). Although CNNs and Transformers are more advanced architectures, the multi-layer perceptron stays a useful baseline for study of deep learning. This is how neural networks learn from data.

A key design consideration in the construction of neural networks is how deep the model should be, where depth is measured by the number of hidden layers lying between the output and input layers. An increase in the number of hidden layers of a neural network allows it to learn more complex representations. However, larger model architectures typically increase the costs of training and inference as well as their optimisation difficulties.

Understanding the relationship between network depth and model performance is therefore an essential part of neural network design. Selecting a proper architecture can significantly affect both training efficiency and predictive accuracy.

This tutorial explores **the impact of hidden layer depth on the performance of Multilayer Perceptron's** using the **Fashion-MNIST dataset,** a widely used benchmark dataset for image classification. Multiple neural network architectures with different hidden layer depths are trained and compared to evaluate how increasing network depth affects learning behaviour, accuracy, and generalisation performance.

The goal of this tutorial is to provide an accessible yet technically grounded explanation of how network depth influences neural network performance, allowing practitioners to better understand how to design effective neural architectures for classification tasks.

2. Multilayer Perceptron's

A Multilayer Perceptron is a type of feedforward neural network consisting of an input layer, one or more hidden layers, and an output layer. Each layer holds neurons that perform linear transformations followed by non-linear activation functions.

For a given neuron, the computation can be expressed as:

$$z = Wx + b$$
$$a = f(z)$$

where:

- W stands for the weights connecting neurons.
- x stands for the input vector.
- b stands for the bias term.
- f is a non-linear activation function?

Some activation functions are: ReLU, sigmoid, tanh. The efficiency and effectiveness of ReLU activation have made it an extremely popular choice in various neural networks nowadays.

The activation function used in output layer depends on nature of task. In multi-class classification problems, the SoftMax function is often used to derive a probability distribution over the classes.

Through backpropagation, which computes the gradient of the loss function relative to each weight, the network weights of an MLP are adjusted during learning. To iteratively update the weights, an optimisation algorithm like stochastic gradient descent (SGD) or Adam is used.

The use of layers in a neural network that do not interact with the outside world is a common way of using hidden layers. Adding more hidden layers to a neural network enhances its capacity to stand for data.

3. Importance of Hidden Layer Depth

The depth of a neural network finds how many layers of feature transformations the input data undergoes before producing an output. Shallow neural networks typically learn simple feature representations, while deeper networks can learn hierarchical structures.
For image classification problems, deeper layers may capture increasingly complex features such as:

- edges and textures
- shapes and patterns
- higher-level object characteristics

However, higher depth does not always lead to better performance. When the networks become too deep, then multiple problems arise.

- A situation called overfitting occurs when the model memorises the training data.
- When gradients become too small to effectively update early layers, this is known as vanishing gradients.
- Increased training time and greater computing cost.

Therefore, finding the best network depth is a major step in neural network design.

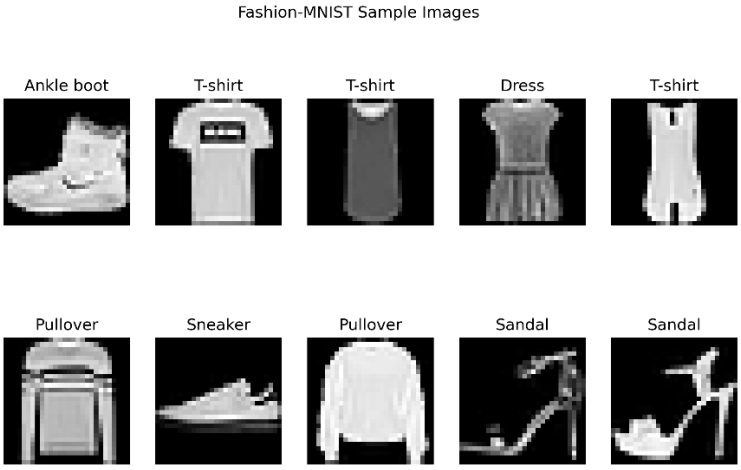
4. Dataset: Fashion-MNIST

The Fashion-MNIST dataset is a benchmark dataset developed by Zalando Research as a more challenging alternative to the traditional MNIST handwritten digit dataset.

The dataset consists of grayscale images being various categories of clothing items.

- T-shirt/top
- Trouser
- Pullover
- Dress
- Coat
- Sandal
- Shirt
- Sneaker
- Bag
- Ankle boot

Each image is represented as a 28 × 28 pixel grayscale image. Before training, pixel values are normalised to the range 0–1 to improve neural network training stability. We can see the below sample image of dataset.



This dataset is particularly suitable for studying neural network behaviour because it provides a moderately challenging classification task without requiring complex models.

5. Experimental Design

We trained and compared several MLPs with varying numbers of hidden layers to investigate the impact of layer depth.

Each hidden layer is composed of 128 neurons which have the ReLU activation function. In the output layer, an activation function SoftMax is used for 10 neurons which denote the 10 classes.

Model	**Hidden Layers**
MLP- 1	1 hidden layer
MLP – 2	2 hidden layers
MLP – 3	3 hidden layers
MLP – 4	4 hidden layers
MLP – 5	5 hidden layers

Training configuration:

Parameter	**Value**
Optimizer	Adam
Loss function	Spares categorical cross – entropy
Epochs	10
Batch Size	128
Validation Split	20%

Multiple visualisations were generated during training to analyse model behaviour, including:

- Training accuracy curves
- Validation accuracy curves
- Training loss curves
- Accuracy comparison across network depths
- Confusion matrix for the best performing model
- Sample prediction visualisations.

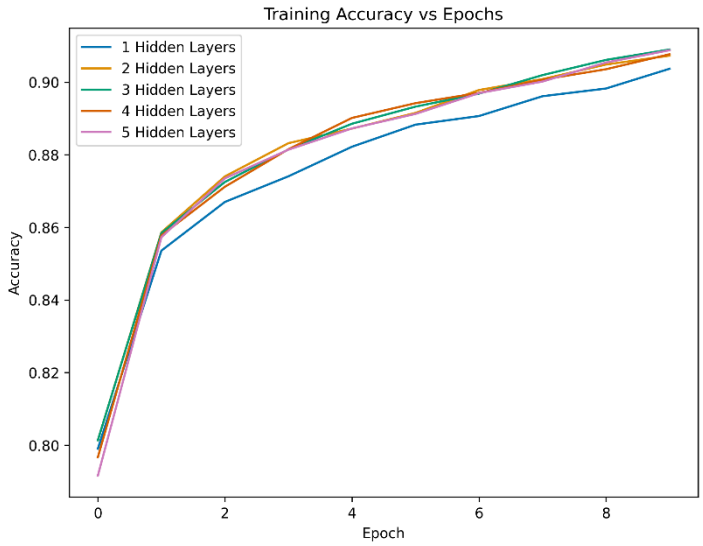
These visualisations help illustrate how network depth influences learning behaviour and classification performance.

6. Results and Analysis

This section presents the experimental results obtained from training Multilayer Perceptron models with different hidden layer depths. The purpose of these experiments is to analyse how increasing the number of hidden layers influences training behaviour, classification accuracy, and overall model performance.

→Training Accuracy Behaviour

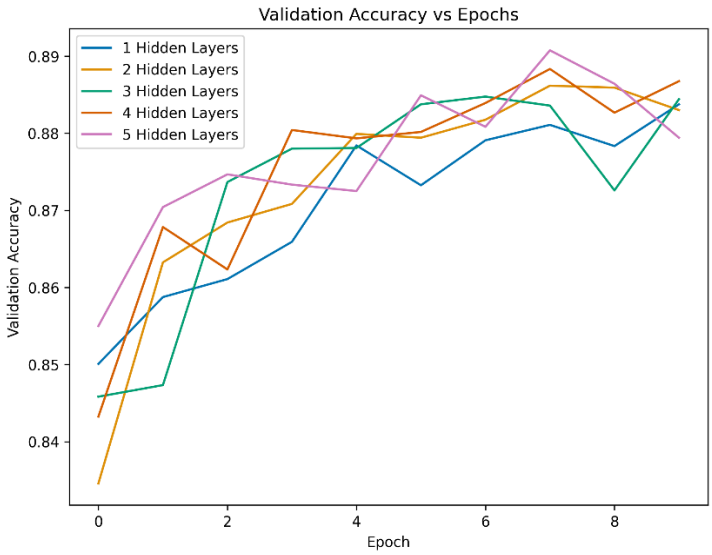
The first experiment examines how training accuracy evolves during the learning process for networks with different hidden layer depths.



The figure proves the training accuracy curves for the neural networks that has 1 to 5 hidden layers. The findings show that deeper networks gain training accuracy faster than shallow networks. Deeper architectures can achieve lower training errors than shallow networks due to their greater representational capacity.

However, high training accuracy alone does not necessarily show good model generalisation. Therefore, validation accuracy must also be analysed.

→Validation Accuracy Behaviour

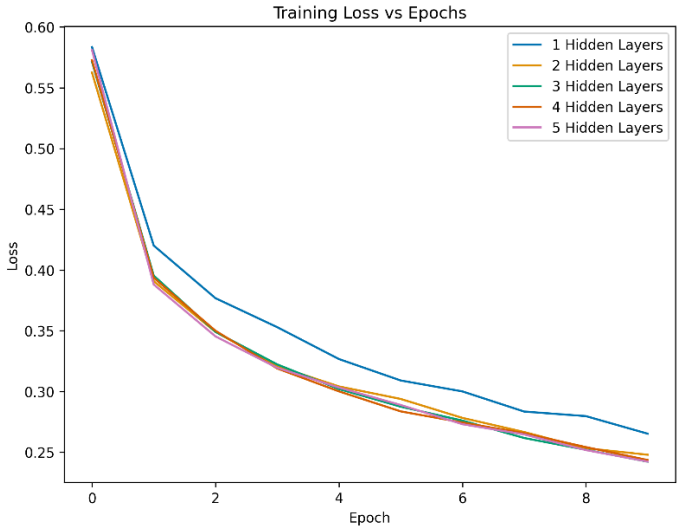


The validation accuracy curves show that models with moderate depth perform well on unseen data. Networks with three or four hidden layers generally achieve strong validation accuracy while keeping stable training behaviour.

In contrast, extremely shallow networks may struggle to capture the complexity of the dataset, resulting in lower validation performance.

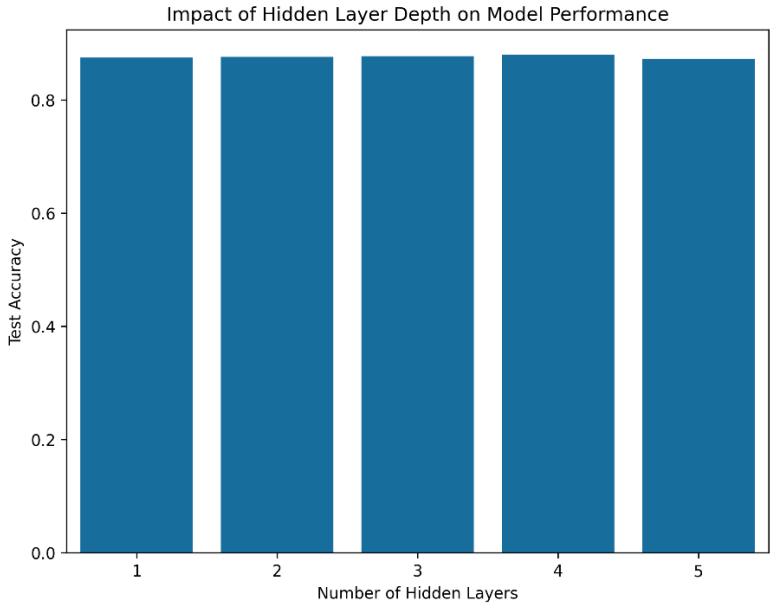
→Training Loss Analysis

Training loss offers more insight into how efficiently the neural networks learn during optimisation.



The training loss curves show how quickly the models reduce classification error during the training process. Deeper networks typically show faster loss reduction because they can learn more complex feature transformations. However, after several epochs the loss values begin to stabilise, showing that the model has reached a reasonable solution.

→Impact of Network Depth on Accuracy

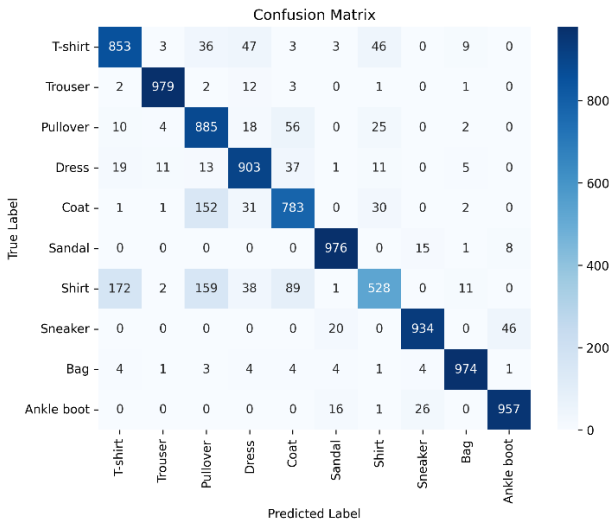


This graph compares the ultimate classification accuracy at different hidden layer depths. Initially, the performance gets better as the network depth increases. Yet, after a given period the improvement gets smaller, which tells of less benefit.

This trend highlights the importance of selecting a proper network architecture rather than simply increasing depth indefinitely.

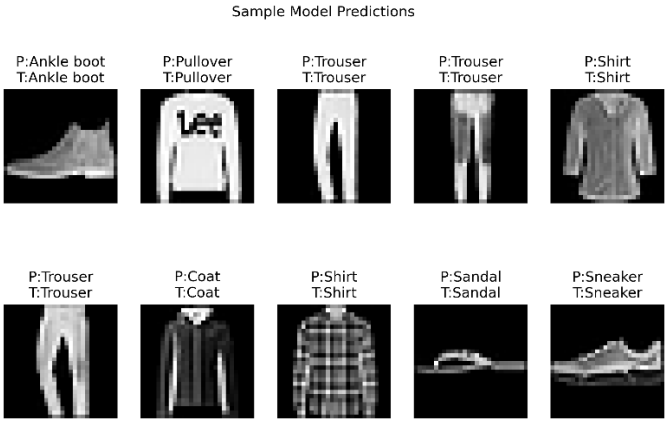
→Confusion Matrix Analysis

The confusion matrix decides how many times an article is classified incorrectly and correctly. Most categories are classified successfully, although some visually similar classes like shirts and coats could be confused sometimes. This kind of analysis gives valuable insight on which categories are difficult for the model.



→Example Predictions

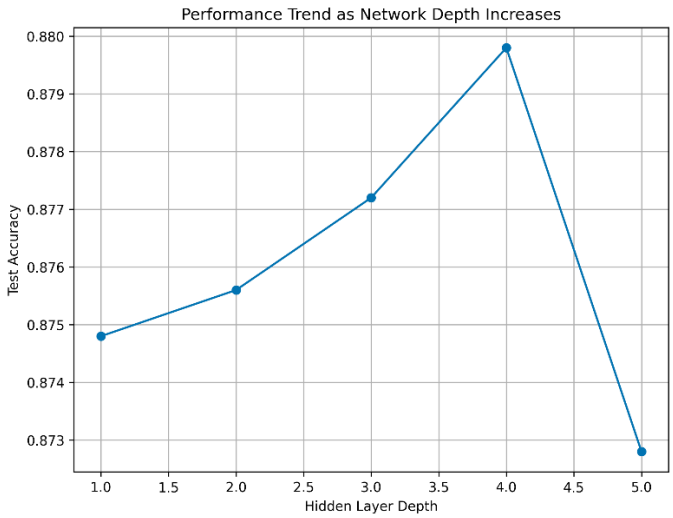
Finally, several example predictions from the trained model are visualised to illustrate real classification outcomes.



Each example shows the predicted label and the true label for a given test image. In most cases, the model correctly finds the clothing item. These examples help prove the practical performance of the neural network on unseen data.

→Performance Trend Across Network Depth

The trend line shows that deeper hidden layers tend to yield better performance until a certain point. After this point, adding more skin layers provides little gain. The observation implies that a moderate network depth will often provide a good trade-off of complexity and performance.



7. Discussion

The experimental results reveal several key takeaways about neural network architecture design.

Shallow neural networks having just a singular hidden layer may not be able the complexity of image classification tasks. While able to learn simple patterns, their representational power is limited.

Moreover, the increased number of hidden layers can considerably improve an algorithm's performance until a certain point. Networks that are moderately deep learn richer hierarchical representations of image features.

However, increasing depth without limit does not always improve performance. More layers are going to up the computing cost and may also introduce optimization problems. The best depth of the network depends on several factors such as the complexity of the dataset, size of training data and computational resources.

8. Conclusion

- Increasing hidden layer depth improves neural network learning capacity.
- Shallow networks may underfit moderately complex datasets.
- Moderate network depths (3–4 layers) typically provide robust performance.
- Added layers beyond a certain point produce diminishing returns.
- Neural network architecture design plays a vital role in classification performance.

Overall, the results prove that carefully selecting neural network depth can significantly improve model performance while keeping efficient training.

9. Future Work

- Examining how the number of neurons in each layer changes the model's overall performance.
- Using dropout and similar measures to improve overfitting.
- Assessing Batch Normalization to Enhance Training Stability
- Comparative analysis of MLP and CNN architecture for Image Classification.
- Examining hyperparameter optimisation techniques for automatic identification of best neural network architectures.

10. References

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